



SIMATS ENGINEERING

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Course Code: ITA1408

Slot: D

Course Name: Ethical Hacking for web Applications

Course Faculty: Dr Samule Rajesh & Dr. K Balamurugan

Project Title: Simulation And Analysis of Intrusion Detection System with Log-Based Forensics

Module Photographs:



Project Description: **STATIC ANALYSIS OF MALWARE**

Module 1 focuses on analyzing suspicious files in a safe and controlled environment without risking the user's actual system. In this module, malware samples are uploaded to Hybrid Analysis (Falcon Sandbox) and OPSWAT MetaDefender, which are industry-standard malware analysis platforms. When a file is submitted, the sandbox executes it inside an isolated virtual environment, where its behavior is carefully observed. The system monitors activities such as file creation, process execution, and system-level changes. This helps in identifying whether the file behaves like a virus. At the same time, OPSWAT MetaDefender performs multi-engine antivirus scanning, where the file is checked using several antivirus engines. This increases detection accuracy by comparing results from multiple sources. To validate the process, the EICAR test file is used. This is a globally accepted test file used only for antivirus testing and does not cause any harm. The tools generate a threat score, risk level, and detailed analysis report, which clearly indicates whether the file is malicious or safe. This module ensures early, accurate, and safe detection of malware, forming the foundation for further analysis in the proposed system.

Student Signature

Guide Signature